

Harshita Kukreja

Email: hk3203@nyu.edu | [🐙 harshitakukreja](https://github.com/harshitakukreja) | [in harshitakukreja8](https://www.linkedin.com/in/harshitakukreja8) | [📷 Harshita Kukreja](https://www.instagram.com/HarshitaKukreja) | Bay Area, CA, USA

EDUCATION

New York University, Courant Institute of Mathematical Sciences
MS Computer Science [GPA: 3.81/4.0]

New York, NY
Sep 2021 – May 2023

Indira Gandhi Delhi Technical University for Women
B.Tech Computer Science and Engineering [Aggregate: 81.3%]

Delhi, India
Aug 2016 – July 2020

EXPERIENCE

Machine Learning Engineer
UCSF Health

San Francisco, CA
May 2024 - Present

- Architected ProgSeg, a general segmentation-based deep learning framework, deploying a customized 3D SegFormer to improve radiation target volume precision over clinical standard-of-care.
- Engineered a multimodal LLM framework to synthesize MRI metadata and 2D slices into structured sequence labels, streamlining the extraction of sequence information from legacy medical archives.
- Developed an automated CCM lesion burden mapping pipeline utilizing a seeded, ML-powered growth algorithm to detect and feature-report small-volume microbleeds, accelerating clinical processing time from hours to minutes.

Software Engineer

University of Maryland Medical Center

Baltimore, MD
Nov 2023 - Feb 2024

- Enhanced the Generalizability of GAN-Based MRI Synthesis (non-FS to FS) across multi-center datasets with a Federated Learning approach achieving 36.9% improvement in SSIM over FastMRI.
- Conducted EDA on large-scale radiology metadata to model the correlation between environmental temperature variations and health disparities.

Machine Learning Research Associate

NYU Langone Health

New York, NY
Sep 2022 - Nov 2023

- Orchestrated a 3D U-Net pipeline (PyTorch Lightning) to synthesize contrast-enhanced MRI volumes from non-contrast scans, achieving a 0.74 SSIM on the NYUMets dataset to preserve structural fidelity while reducing chemical agent dependency.

Software Engineer Intern

Tech For Good Inc.

Boston, MA
Jun 2022 - Aug 2022

- Instrumented the LeCAR **Machine Learning-based caching system** on the org's Mission Uplink platform increasing cache-hit ratio by 10% reducing network latency and enhancing internet connectivity for underserved communities.

Machine Learning Research Associate

Netaji Subhas University of Technology

Delhi, India
Jun 2019 - Aug 2021

- Optimized **denoising** on iris and palmprints by reducing Non-Local Means filter **time by 75%**, achieving PSNR up to 40.86 and 98.39% classification accuracy with a pretrained ResNet50 in PyTorch.
- Developed DeepCrypt, integrating **CNNs with cryptographic hashing** to enable secure cloud storage of biometric templates, achieving 99.56% authentication accuracy.
- Built a multimodal biometric authentication system using **feature fusion** leveraging iris and face modalities orchestrated on **TensorFlow** and **Keras**, achieving 99.8% accuracy while mitigating security limitations.

PROJECTS

Semi Supervised DBSCAN

- Open-sourced a Semi-Supervised version of the unsupervised DBSCAN algorithm using Prim's algorithm. [\[GitHub\]](#)

Promotions Resource in an E-Commerce Website

- Designed the back-end for the Promotions team of an eCommerce website as a collection of **REST API** services for a client.
- Worked on **Docker** containers with **Kubernetes CI/CD** pipelines to integrate code with **nosetests** before deploying it, simulating an Agile development with biweekly sprints.

Autism Spectrum Disorder Screening using Predictive Analytics

- Performed **EDA** on the Autism dataset using **Tableau**, **R**, **Pandas**, and **Matplotlib**
- Applied Logistic Regression, Random Forest, and Decision Tree models, with Random Forest achieving a **99% F1-score**

SKILLS

Languages
ML

Python, Java, C++, C, R, SQL, HTML, CSS, Shell Scripting
PyTorch, PyTorchLightning, TensorFlow, Keras, Jupyter, NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, Tableau, pyalfe, Vision Transformers, Multi-modal LLM

Software

Linux, Git, Agile, CI/CD, Docker, Kubernetes, SQLAlchemy, A/B Testing, IBM Cloud

PUBLICATIONS

ProgSeg: A General Segmentation-based Framework for Glioblastoma Radiotherapy Planning. *Harshita Kukreja et al.* — *Computer Vision and Pattern Recognition (CVPR AI4RWC Workshop)* 2026

Personalizing Radiation Planning Using a Transformer-Based Segmentation Approach to Predict Glioblastoma Recurrence from Metabolic MRI and White Matter Infiltration Maps. Harshita Kukreja et al. — <i>American Society of Neuroradiology</i> (Abstract)	2026
Detection in Progress - A Multimodal Segmentation-based Approach for Predicting Glioblastoma Recurrence. Harshita Kukreja et al. — <i>Neural Information Processing Systems (NeurIPS Imageomics Workshop)</i>	2025
Transformer-Based Approach to Predict Glioblastoma Recurrence from MRSI and Diffusion MRI for Personalizing Radiation Planning. Harshita Kukreja et al. — <i>Society for Neuro-Oncology</i> (Abstract)	2025
Defining Radiation Target Volumes with AI-Driven Predictions of Glioma Recurrence from MRSI, Diffusion MRI, and Transformers. Harshita Kukreja et al. — <i>International Society for Magnetic Resonance in Medicine</i> (Abstract)	2025
Large Language Model Based Identification of Brain MRI Sequences. Radhika Bhalerao, Harshita Kukreja et al. — <i>International Society for Magnetic Resonance in Medicine</i> (Abstract)	2025
Longitudinal voxel-wise analysis using a novel deep-learning-derived pKI-67 for early prediction of glioblastoma patients prognosis. Nate Tran, Harshita Kukreja et al. — <i>International Society for Magnetic Resonance in Medicine</i> (Abstract)	2025
Improving Multi-Center Generalizability of GAN-Based Fat Suppression using Federated Learning. Pranav Kulkarni, Adway Kanhere, Harshita Kukreja et al. — <i>Medical Imaging with Deep Learning</i> [Link]	2024
Artificial Intelligence Analysis of Social Media Posts on Glucagon-Like-Peptide-1 Receptor Agonists (GLP-1RA). Aamir Javaid, Harshita Kukreja et al. — <i>Journal of the American College of Cardiology</i> [Link]	2024
An Evaluation of Denoising Techniques and Classification of Biometric Images Based on Deep Learning. Shefali Arora, Ruchi Mittal, Harshita Kukreja et al. — <i>Multimedia Tools and Applications</i> , Springer [Link]	2022
Privacy Enhancement in Biometric Systems by Template Protection using Deep Learning. Harshita Kukreja et al. — <i>International Journal of Scientific Research in Computer Science and Engineering</i> [Link]	2021
Privacy Protection of Biometric Templates Using Deep Learning. Shefali Arora, M.P.S. Bhatia, Harshita Kukreja et al. — <i>Innovations in Cyber Physical Systems</i> , Springer [Link]	2021
A Multimodal Biometric System for Secure User Identification Based on Deep Learning. Shefali Arora, M.P.S. Bhatia, Harshita Kukreja — <i>International Conference on Information and Communication Technology</i> , Springer [Link]	2021

ACADEMIC SERVICE

Reviewer

Medical Image Computing and Computer Assisted Intervention [Link]	2026
Journal of Imaging Informatics in Medicine [Link]	2026
CVPR Women in CV Workshop [Link]	2026
NeurIPS Imageomics Workshop [Link]	2025

Teaching Assistant

Handled grading of assignments and tutoring of students, created and hosted hands-on exercises, and feedback-sessions.

DS-UA 301 — Advanced Topics in ML/DL (Prof. Parijat Dube)	Spring 2023
CSCI-UA 472 — Artificial Intelligence (Prof. Ernest Davis)	Spring 2023
CSCI-UA 310 — Basic Algorithms (Prof. Vladimir Podolskii)	Spring 2023
CSCI-UA 310 — Basic Algorithms (Prof. Marshall Ball)	Fall 2022
CSCI-GA 2271 — Computer Vision (Prof. Rob Fergus)	Fall 2022
CSCI-GA 3033 — Introduction to Deep Learning Systems (Prof. Parijat Dube)	Fall 2022